Ethical Hacking Lab 1

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PGP (Pretty Good Privacy)

Created users Sam, Jack and Mike

```
[root@localhost cent]# useradd sam
[root@localhost cent]# useradd jack
[root@localhost cent]# useradd mike
```

Provided password to each users

```
[root@localhost cent]# passwd sam
Changing password for user sam.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost cent]# passwd jack
Changing password for user jack.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost cent]# passwd mike
Changing password for user mike.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost cent]#
```

Create group Sales

```
[root@localhost ~]# groupadd sales
```

Added all users in Sales group

```
[root@localhost ~]# usermod -a -G sales sam
[root@localhost ~]# usermod -a -G sales jack
[root@localhost ~]# usermod -a -G sales mike
[root@localhost ~]#
```

Created directory Sales

```
[root@localhost ~]# mkdir /sales
[root@localhost ~]#
```

Checked the permissions in the Sales group and allowed rwx permission

```
[root@localhost /]# ll / | grep sales
drwxr-xr-x.   2 root root    6 Oct 12 22:42 sales
```

```
[root@localhost /]# chgrp sales /sales
[root@localhost /]# ll / | grep sales
drwxr-xr-x.   2 root sales    6 Oct 12 22:42 sales
```

Generate Sam’s encryption key through gpg

```
[sam@localhost ~]$ gpg --full-generate-key
gpg (GnuPG) 2.3.3; Copyright (C) 2021 Free Software Foundation, Inc.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:
   (1) RSA and RSA (default)
   (2) DSA and Elgamal
   (3) DSA (sign only)
   (4) RSA (sign only)
  (14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
         0 = key does not expire
      <n>  = key expires in n days
      <n>w = key expires in n weeks
      <n>m = key expires in n months
      <n>y = key expires in n years
Key is valid for? (0) 1y
Key expires at Tue 12 Oct 2027 10:52:02 PM IST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: SAM
Name must be at least 5 characters long
Real name: Sammy
Email address: sam@mail.com
Comment:
```

```
Comment:
You selected this USER-ID:
    "Sammy <sam@mail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/sam/.gnupg/trustdb.gpg: trustdb created
gpg: key AAA8C078304AA68D marked as ultimately trusted
gpg: directory '/home/sam/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/sam/.gnupg/openpgp-revocs.d/
public and secret key created and signed.

pub   rsa4096 2026-10-12 [SC] [expires: 2027-10-12]
      04AE4FE220DAA1EF6ED54872AAA8C078304AA68D
uid                      Sammy <sam@mail.com>
sub   rsa4096 2026-10-12 [E] [expires: 2027-10-12]

[sam@localhost ~]$
```

Export Sam's encryption key as **.asc** file

```
[sam@localhost ~]$ cd .gnupg/
[sam@localhost .gnupg]$ ll
total 12
drwx------. 2 sam sam   58 Oct 12 22:52 openpgp-revocs.d
drwx------. 2 sam sam  110 Oct 12 22:52 private-keys-v1.d
-rw-r--r--. 1 sam sam 2478 Oct 12 22:52 pubring.kbx
-rw-------. 1 sam sam   32 Oct 12 22:51 pubring.kbx~
-rw-------. 1 sam sam 1240 Oct 12 22:52 trustdb.gpg
[sam@localhost .gnupg]$ gpg --armor --export sam@mail.com > sam_pub.asc
[sam@localhost .gnupg]$ ll
total 16
drwx------. 2 sam sam   58 Oct 12 22:52 openpgp-revocs.d
drwx------. 2 sam sam  110 Oct 12 22:52 private-keys-v1.d
-rw-r--r--. 1 sam sam 2478 Oct 12 22:52 pubring.kbx
-rw-------. 1 sam sam   32 Oct 12 22:51 pubring.kbx~
-rw-r--r--. 1 sam sam 3134 Oct 12 22:54 sam_pub.asc
-rw-------. 1 sam sam 1240 Oct 12 22:52 trustdb.gpg
```

Generate Jake's encryption key through gpg

```
[jack@localhost ~]$ gpg --full-generate-key
gpg (GnuPG) 2.3.3; Copyright (C) 2021 Free Software Foundation, Inc.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

gpg: directory '/home/jack/.gnupg' created
gpg: keybox '/home/jack/.gnupg/pubring.kbx' created
Please select what kind of key you want:
   (1) RSA and RSA (default)
   (2) DSA and Elgamal
   (3) DSA (sign only)
   (4) RSA (sign only)
  (14) Existing key from card
Your selection?
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072)
Requested keysize is 3072 bits
Please specify how long the key should be valid.
         0 = key does not expire
      <n>  = key expires in n days
      <n>w = key expires in n weeks
      <n>m = key expires in n months
      <n>y = key expires in n years
Key is valid for? (0) 1y
```

```
Comment:
You selected this USER-ID:
    "Jackie <jack@mail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/jack/.gnupg/trustdb.gpg: trustdb created
gpg: key 814CD4ADDE056EE8 marked as ultimately trusted
gpg: directory '/home/jack/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/jack/.gnupg/openpgp-revocs.d/
EE8.rev'
public and secret key created and signed.

pub   rsa3072 2026-10-12 [SC] [expires: 2027-10-12]
      AB60991D06445360F8068393814CD4ADDE056EE8
uid                      Jackie <jack@mail.com>
sub   rsa3072 2026-10-12 [E] [expires: 2027-10-12]
```

Copied and imported Sam's public key file (as Jack)

```
[jack@localhost ~]$ cp /tmp/sam_pub.asc .
[jack@localhost ~]$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  sam_pub.asc  Te
[jack@localhost ~]$ gpg --import sam_pub.asc
gpg: key AAA8C078304AA68D: public key "Sammy <sam@mail.com>" imported
gpg: Total number processed: 1
gpg:               imported: 1
[jack@localhost ~]$
```

Generate Mike's encryption key through gpg

```
[mike@localhost ~]$ gpg --full-generate-key
gpg (GnuPG) 2.3.3; Copyright (C) 2021 Free Software Foundation, Inc.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

gpg: directory '/home/mike/.gnupg' created
gpg: keybox '/home/mike/.gnupg/pubring.kbx' created
Please select what kind of key you want:
   (1) RSA and RSA (default)
   (2) DSA and Elgamal
   (3) DSA (sign only)
   (4) RSA (sign only)
  (14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
         0 = key does not expire
      <n>  = key expires in n days
      <n>w = key expires in n weeks
      <n>m = key expires in n months
      <n>y = key expires in n years
Key is valid for? (0) 1y
Key expires at Tue 12 Oct 2027 11:04:25 PM IST
Is this correct? (y/N) y
```

```
GnuPG needs to construct a user ID to identify your key.

Real name: Micheal
Email address: mike@mail.com
Comment:
You selected this USER-ID:
    "Micheal <mike@mail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/mike/.gnupg/trustdb.gpg: trustdb created
gpg: key 7BFE71915862E139 marked as ultimately trusted
gpg: directory '/home/mike/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/mike/.gnupg/openpgp-revocs.d/D
public and secret key created and signed.

pub   rsa4096 2026-10-12 [SC] [expires: 2027-10-12]
      D884201E03F2D1C22EFB9C0D7BFE71915862E139
uid                      Micheal <mike@mail.com>
sub   rsa4096 2026-10-12 [E] [expires: 2027-10-12]

[mike@localhost ~]$
```

Copied and imported Sam's public key file (as Mike)

```
[mike@localhost ~]$ cp /tmp/sam_pub.asc .
[mike@localhost ~]$ ll
total 4
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Desktop
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Documents
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Downloads
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Music
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Pictures
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Public
-rw-r--r--. 1 mike mike 3134 Oct 12 23:05 sam_pub.asc
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Templates
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Videos
[mike@localhost ~]$ gpg --import sam_pub.asc
gpg: key AAA8C078304AA68D: public key "Sammy <sam@mail.com>" imported
gpg: Total number processed: 1
gpg:               imported: 1
[mike@localhost ~]$
```

As Mike, created sales data file and encrypted it with Sam's public key and copied it to the Sales directory.

```
[mike@localhost ~]$ vim sales_data_mike.txt
[mike@localhost ~]$ gpg --encrypt -r sam@mail.com sa
sales_data_mike.txt  sam_pub.asc
[mike@localhost ~]$ gpg --encrypt -r sam@mail.com sales_data_mike.txt
gpg: 6246E39936FEA36B: There is no assurance this key belongs to the named user

sub  rsa4096/6246E39936FEA36B 2026-10-12 Sammy <sam@mail.com>
 Primary key fingerprint: 04AE 4FE2 20DA A1EF 6ED5  4872 AAA8 C078 304A A68D
      Subkey fingerprint: 3BA7 2158 10D4 42EE AB74  25C1 6246 E399 36FE A36B

It is NOT certain that the key belongs to the person named
in the user ID.  If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) Y
[mike@localhost ~]$ ll
total 12
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Desktop
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Documents
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Downloads
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Music
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Pictures
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Public
-rw-r--r--. 1 mike mike   19 Oct 12 23:07 sales_data_mike.txt
-rw-r--r--. 1 mike mike  630 Oct 12 23:08 sales_data_mike.txt.gpg
-rw-r--r--. 1 mike mike 3134 Oct 12 23:05 sam_pub.asc
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Templates
drwxr-xr-x. 2 mike mike    6 Oct 12 23:03 Videos
[mike@localhost ~]$ cp sales_data_mike.txt.gpg /sales/
[mike@localhost ~]$
```

```
[mike@localhost ~]$ ll /sales/
total 4
-rw-r--r--. 1 mike mike 630 Oct 12 23:08 sales_data_mike.txt.gpg
[mike@localhost ~]$
```

As Jack, created sales data file and encrypted it with Sam's public key and copied it to the Sales directory.

```
[jack@localhost ~]$ vim sales_data_jack.txt
[jack@localhost ~]$ gpg --encrypt -r sam@mail.com sales_data_jack.txt
gpg: 6246E39936FEA36B: There is no assurance this key belongs to the named user

sub  rsa4096/6246E39936FEA36B 2026-10-12 Sammy <sam@mail.com>
 Primary key fingerprint: 04AE 4FE2 20DA A1EF 6ED5  4872 AAA8 C078 304A A68D
      Subkey fingerprint: 3BA7 2158 10D4 42EE AB74  25C1 6246 E399 36FE A36B

It is NOT certain that the key belongs to the person named
in the user ID.  If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
[jack@localhost ~]$ cp sales_data_jack.txt.gpg /sales/
```

Verified that both files are in Sales directory

```
[jack@localhost ~]$ ll /sales/
total 8
-rw-r--r--. 1 jack jack 629 Oct 12 23:13 sales_data_jack.txt.gpg
-rw-r--r--. 1 mike mike 630 Oct 12 23:08 sales_data_mike.txt.gpg
[jack@localhost ~]$
```

As Sam, decrypted both the files into his home directory

```
[sam@localhost ~]$ ll /sales/
total 8
-rw-r--r--. 1 jack jack 629 Oct 12 23:13 sales_data_jack.txt.gpg
-rw-r--r--. 1 mike mike 630 Oct 12 23:08 sales_data_mike.txt.gpg
[sam@localhost ~]$ gpg --decrypt /sales/sales_data_jack.txt.gpg > sales_data_jack.txt
gpg: encrypted with rsa4096 key, ID 6246E39936FEA36B, created 2026-10-12
      "Sammy <sam@mail.com>"
```

```
[sam@localhost ~]$ gpg --decrypt /sales/sales_data_mike.txt.gpg > sales_data_mike.txt
gpg: encrypted with rsa4096 key, ID 6246E39936FEA36B, created 2026-10-12
      "Sammy <sam@mail.com>"
[sam@localhost ~]$
```

Verified the procedure and decrypted the files successfully

```
[sam@localhost ~]$ ll
total 8
drwxr-xr-x. 2 sam sam  6 Oct 12 22:50 Desktop
drwxr-xr-x. 2 sam sam  6 Oct 12 22:50 Documents
drwxr-xr-x. 2 sam sam  6 Oct 12 22:50 Downloads
drwxr-xr-x. 2 sam sam  6 Oct 12 22:50 Music
drwxr-xr-x. 2 sam sam  6 Oct 12 22:50 Pictures
drwxr-xr-x. 2 sam sam  6 Oct 12 22:50 Public
-rw-r--r--. 1 sam sam 19 Oct 12 23:15 sales_data_jack.txt
-rw-r--r--. 1 sam sam 19 Oct 12 23:16 sales_data_mike.txt
drwxr-xr-x. 2 sam sam  6 Oct 12 22:50 Templates
drwxr-xr-x. 2 sam sam  6 Oct 12 22:50 Videos
[sam@localhost ~]$ cat sales_data_jack.txt
Jack's sales data.
[sam@localhost ~]$ cat sales_data_mike.txt
Mike's sales data.
[sam@localhost ~]$
```